25. If $\omega = cis\frac{2\pi}{3}$, then the number of distinct roots of $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$

(1) 1 (2) 2 (3) 3 (4) 4

## SUMMARY

In this chapter we studied

**Rectangular form of a complex number is $x+iy$ (or $x+yi$), where $x$ and $y$ are real numbers.**

**Two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are said to be equal if and only if $\text{Re}(z_1) = \text{Re}(z_2)$ and $\text{Im}(z_1) = \text{Im}(z_2)$. That is $x_1 = x_2$ and $y_1 = y_2$.**

**The conjugate of the complex number $x+iy$ is defined as the complex number $x-iy$.**

Properties of complex conjugates

(1) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

(2) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$

(3) $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$

(4) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$, $z_2 \neq 0$

(5) $\text{Re}(z) = \frac{z+\overline{z}}{2}$

(6) $\text{Im}(z) = \frac{z-\overline{z}}{2i}$

(7) $\overline{(z^n)} = (\overline{z})^n$, where $n$ is an integer

(8) $z$ is real if and only if $z = \overline{z}$

(9) $z$ is purely imaginary if and only if $z = -\overline{z}$

(10) $\overline{\overline{z}} = z$

**If $z = x+iy$, then $\sqrt{x^2+y^2}$ is called modulus of $z$. It is denoted by $|z|$.**

Properties of Modulus of a complex number

(1) $|z| = |\overline{z}|$

(2) $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle inequality)

(3) $|z_1 z_2| = |z_1||z_2|$

(4) $|z_1 - z_2| \geq ||z_1| - |z_2||$

(5) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$, $z_2 \neq 0$

(6) $|z^n| = |z|^n$, where $n$ is an integer

(7) $\text{Re}(z) \leq |z|$

(8) $\text{Im}(z) \leq |z|$



Chapter 2 Complex Numbers.indd 95 31-01-2020 17:38:20